

H-LAYER closure – Final Consolidation: the complete T-016 to T-024

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

This note is the single canonical record of the H-LAYER closure programme. It states the proposition, the decomposition, the complete proof chain T-016 to T-024, the two-tier competitor-class decision, and the honest standing of B1/B2. It introduces no new result; every number is cited to its step note + script and re-verified end-to-end by `res5_025_hlayer_final_consolidation.py` (7/7).

2. The proposition

After the SC-SCOPE lift (2026-06-09), B1-RH-ENUM (the Reading-H enumerated selection) and B2-PROPA-HLAYER (Proposition A) rest on the SOLE hypothesis

H-LAYER: the isotropic Gaussian–Hartree variational layer is the comparison infimum,

i.e. no admissible competitor lowers the free energy below the isotropic dressing $G_*(q) = 1/(r_R + c(|q|^2 - q_0^2)^2)$. The open content was STEP-5B: the beyond-diagonal, beyond-second-cumulant refinements for non-enumerated patterns.

3. Decomposition

The comparison splits into three sectors: (D) diagonal – is G_* the diagonal-class minimum (Math427/H-diag)? (O) off-diagonal – can a Bloch covariance lower F (RES-1)? (S) selection floor – does the STEP-5B layer-margin survive the third-cumulant endpoint correction (SC-SCOPE)?

4. The complete proof chain (T-016 to T-024)

T-016 (D) The condensate-free Hessian is BLOCK-DIAGONAL in angular momentum: the rotation-invariant interaction curves only $l = 0$ ($\Phi''_{\text{diag}} = \frac{3}{2}u_{\text{eff}} = 4.03 > 0$); every $l \geq 1$ sees the entropy Hessian $\frac{1}{2}G_*^{-2} \geq \frac{1}{2}r_R^2 = 4.64 \times 10^{-2} > 0$ (ANALYTIC, at the gap shell). G_* is a STRICT curvature-protected diagonal minimum.

T-017 (S) Weighted Lemma A (Cauchy–Schwarz, $r(t) \leq T'$): $K_{\text{floor}}(c) \leq T'(Q)$ for ANY amplitudes ($\chi(P)$ bypassed). Registered pin $T' \leq 13 < 20.5 < 26.2$.

T-018, T-019 (O) B^{cond} is diagonal in Q , so $\|B^{\text{cond}}\| = R_{\text{lead}}(Q) \leq 0.650$ on the registered class. For TECT's LOCAL interaction there is NO A-independent attractive Fock exchange: the off-diagonal mean-field Hessian is the stabilising density-density term $+\frac{3}{2}u_{\text{eff}} > 0$ (Hartree=Fock). The genuine residual is the A-dependent bubble (R_{lead}) + the two-loop = SC-SCOPE.

T-020 (O) The Bogoliubov band is positive iff $\rho R_{\text{lead}} < 1$; at $\rho = 1$ (no resummation credit), $R_{\text{lead}}(13) = 0.650 < 1$ – second-cumulant off-diagonal stability CLASS-WIDE on the registered class.

T-021 (S) MARGIN cancels in the certified joint: $\text{joint}(\rho_{\text{lat}})$ is monotone, $\text{joint} > 1 \iff \rho_{\text{lat}} > 4.347 \iff T' < 60.4$. Registered $T' \leq 13 < 60.4$ gives $\text{joint}(13) = \times 1.097 > 1$ – SC-SCOPE third-cumulant endpoint CLASS-WIDE.

, **T-023**, **T-024** Competitor-class: consolidation + four exclusion boundaries + the operator decision (Sec.5).

5. The two-tier competitor-class decision (operator)

The sharp pin $T' \leq 13$ is class-specific; an adversarial lattice-but-off-shell escape ($N = 40$, $T' = 20 > 13$) exists. The operator decision (offshell-operator-decision v1.0) splits the class:

$$\mathcal{A}_{\text{adm}} = \{\text{registered crystalline shell/shell-union}\} (T' \leq 13),$$

$$\mathcal{A}_{\text{ext}} = \{Q = -Q, \#Q \leq n_{\text{pack}}\} (T' \leq \frac{N}{2} \leq 20).$$

Off-shell is excluded from the primary class by the kinetic penalty: $K_0(k) = r_R + c(|k|^2 - q_0^2)^2$ is minimised on-shell, and an off-shell mode at $|k| = \sqrt{2}q_0$ costs $K_0 - r_R = 0.214 = \times 50$ the selection MARGIN – an off-shell mode cannot be a leading Reading-H competitor *unless its interaction-energy gain compensates the kinetic penalty*, $\Delta F_{\text{int}}^{\text{gain}} > \Delta F_{\text{kin}}^{\text{offshell}}$ (the falsification gate). The universal antipodal fallback $T' \leq N/2$ (sum-circles C_t, C_{-t} disjoint for $t \neq 0$, $2T' \leq N$) keeps $\mathcal{A}_{\text{ext}}$ controlled.

6. Final status (both tiers below every threshold)

	T'	$R_{\text{lead}} (< 1?)$	$\text{joint} (> 1?)$	verdict
Primary $\mathcal{A}_{\text{adm}}$	≤ 13	$0.650 (\times 1.54)$	$\times 1.097$	comfortable
Adversarial $\mathcal{A}_{\text{ext}}$	≤ 20	$0.974 (\times 1.026)$	$\times 1.082$	razor-thin, unbroken
thresholds		RES-1 20.5	RES-5 26.2	SC-SCOPE 60.4

Both $T' \leq 13$ and $T' \leq 20$ lie below $20.5 < 26.2 < 60.4$: the H-LAYER analytic closure holds on the primary class comfortably and on the adversarial class thinly.

7. Standing of B1/B2 (honest ceiling)

No tier flip is enacted. B1/B2 remain **T6 on {H-LAYER}**, B5 T5. The available move, on operator sign-off of the shell-supported primary class, is a NAMED-HYPOTHESIS-REDUCED T6: the conditioning hypothesis changes from “H-LAYER” to “shell-supported crystalline completeness”, and the proof inherits – not eliminates – the thin-certified grades: $R_{\text{max}} = 0.385$ (the Parseval-pinned quartic), the sunset cap $S = 1.13$, the T-017 pin, and the anchoring ($u_{\text{eff}} > 0$, $g \geq 0$ on $[\times 0.5, \times 2]$). These are STRONG-EVIDENCE / thin-certified, NOT unconditional theorems; hence the honest ceiling is hypothesis-reduced T6, not T7.

8. What remains (operator-optional)

(i) Enact the B1/B2 re-tier to hypothesis-reduced T6 on the shell-supported class; or (ii) harden the inherited thin-certified grades (the R_{max} certificate, the sunset cap, the anchoring band) for a stronger ceiling. No analytic H-LAYER gap remains.

9. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “A consolidation that claims closure.” CALIBRATED: it records the ANALYTIC closure on the registered class + the adversarial fallback + the operator competitor decision. It does NOT flip B1/B2 and ceilings any re-tier at hypothesis-reduced T6.
- (β) “The adversarial RES-1 margin is $\times 1.026$.” ACKNOWLEDGED and recorded in the status box; the comfortable margins are the primary (shell-supported) class.
- (γ) “Inherited thin grades are load-bearing.” YES, and §7 names them explicitly as the reason the ceiling is T6, not T7.

Quantitative sanity check (mandatory): end-to-end re-verification (7/7): $\Phi''_{\text{diag}} = 4.03 > 0$, entropy $4.64 \times 10^{-2} > 0$; $13 < 20.5 < 26.2$; $R_{\text{lead}}(13) = 0.650 < 1$; $\text{joint}(13) = \times 1.097 > 1$ ($T'_{\text{crit}} = 60.4$); off-shell penalty $\times 50$ MARGIN; $R_{\text{lead}}(20) = 0.974 < 1$, $\text{joint}(20) = \times 1.082 > 1$; $20 < 20.5 < 26.2 < 60.4$.

10. Result footer

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Result ID:          B2-PROPA-HLAYER / H-LAYER closure Final
                    Consolidation
Precise statement:  The T-016..T-024 programme closes every
                    ANALYTIC H-LAYER residual
                    class-wide on the primary shell-supported class
                    A_adm (T'<=13:
                    R_lead=0.650, joint=x1.097), and the analytic
                    thresholds survive on
                    the adversarial class A_ext (T'<=N/2<=20:
                                                    R_lead=0.974,
                                                    joint=x1.082);
                    both below RES-1 20.5 < RES-5 26.2 < SC-SCOPE
                    60.4. Off-shell is
                    excluded from A_adm by the kinetic penalty (x50
                    MARGIN per off-shell
                    mode), retained in A_ext under T'<=N/2. Sectors:
                                                    (D)
                                                    diagonal
                                                    isotropy

                    strict minimum (T-016); (O) no A-independent
                    Fock exchange + 2nd-
                    cumulant stability (T-018/019/020); (S) chi(P)
                    floor + SC-SCOPE 3rd
                    cumulant (T-017/021). Competitor-class
                    operator-decided (T-022/023/024).
Scope:             Operating endpoint I=2e-3, ROBUSTNESS-MU2
                    [x0.5,x2]. Primary =
                    comfortable; adversarial = thin. Inherits
                    R_max=0.385, sunset 1.13,
                    T-017 pin, anchoring.
Dependencies:      isotropy-infimum-core v1.1;
                    res5-chi-link-bypass v1.1;
                    hdiag-offdiag-gershgorin-threshold v1.1;
                    offdiag-exchange-scalar-
                    identification v1.0;
                    classwide-secondcumulant-stability v1.0;
                    scscope-classwide-endpoint v1.0;
                    hlayer-analytic-closure-consolidation
                    v1.1; aadm-exclusion-boundaries v1.1;
                    offshell-operator-decision v1.0.

```

Evidence grade: FINAL CONSOLIDATION + EXECUTED end-to-end
(res5_025_hlayer_final_consolidation.py 7/7) +
per-step scripts (3-7/each).

Reproduction command: python
codes/vacuum/res5_025_hlayer_final_consolidation.py

Expected output: phi''=4.03; 13<20.5<26.2; R_lead(13)=0.650;
joint(13)=x1.097
(crit T'=60.4); penalty x50; R_lead(20)=0.974;
joint(20)=x1.082; 7/7.

Falsification gate: A re-opened analytic sector (D/O/S); or an
off-shell competitor whose
total free energy (kinetic penalty included)
beats the isotropic layer
($\Delta F_{\text{int}}^{\text{gain}} > \Delta F_{\text{kin}}^{\text{offshell}}$); or
an A_ext set with $T' > N/2$.

Tier before / after: No tier flip. B1/B2 T6 on {H-LAYER}, B5 T5.
Analytic residuals closed
(two-tier); competitor-class operator-decided.
Available ceiling on
sign-off = hypothesis-reduced T6 (H-LAYER ->
shell-supported-crystalline
+ inherited thin grades), NOT unconditional T7.

No-overclaim statement: Does NOT flip B1/B2 and does NOT claim T7.
Records the analytic closure
on the primary class (comfortable) +
adversarial class (thin) + the
operator competitor decision. The
thin-certified grades (R_max, sunset,
anchoring) and the shell-supported completeness
sign-off are inherited,
not eliminated. B1/B2 stay T6 on {H-LAYER}.

Next required action: OPERATOR (optional): enact the
hypothesis-reduced-T6 re-tier on the
shell-supported class, or harden the inherited
thin-certified grades.
No analytic H-LAYER gap remains.